

3.0 I/O PORTS

There are as many as thirty-five general purpose I/O pins available. Depending on which peripherals are enabled, some or all of the pins may not be available as general purpose I/O. In general, when a peripheral is enabled, the associated pin may not be used as a general purpose I/O pin.

3.1 PORTA and the TRISA Registers

PORTA is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISA (Register 3-2). Setting a TRISA bit (= 1) will make the corresponding PORTA pin an input (i.e., disable the output driver). Clearing a TRISA bit (= 0) will make the corresponding PORTA pin an output (i.e., enables output driver and puts the contents of the output latch on the selected pin). Example 3-1 shows how to initialize PORTA.

Reading the PORTA register (Register 3-1) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write

operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISA register (Register 3-2) controls the PORTA pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISA register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

Note: The ANSEL register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

EXAMPLE 3-1: INITIALIZING PORTA

```
BANKSEL PORTA      ;
CLRF   PORTA        ;Init PORTA
BANKSEL ANSEL       ;
CLRF   ANSEL         ;digital I/O
BANKSEL TRISA       ;
MOVLW  0Ch          ;Set RA<3:2> as inputs
MOVWF  TRISA         ;and set RA<5:4,1:0>
                      ;as outputs
```

REGISTER 3-1: PORTA: PORTA REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **RA<7:0>**: PORTA I/O Pin bit
 1 = Port pin is > V_{IH}
 0 = Port pin is < V_{IL}

REGISTER 3-2: TRISA: PORTA TRI-STATE REGISTER

R/W-1 ⁽¹⁾	R/W-1 ⁽¹⁾	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **TRISA<7:0>**: PORTA Tri-State Control bit
 1 = PORTA pin configured as an input (tri-stated)
 0 = PORTA pin configured as an output

Note 1: TRISA<7:6> always reads '1' in XT, HS and LP Oscillator modes.

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3.2 Additional Pin Functions

RA0 also has an Ultra Low-Power Wake-up option. The next three sections describe these functions.

3.2.1 ANSEL REGISTER

The ANSEL register (Register 3-3) is used to configure the Input mode of an I/O pin to analog. Setting the appropriate ANSEL bit high will cause all digital reads on the pin to be read as '0' and allow analog functions on the pin to operate correctly.

The state of the ANSEL bits has no affect on digital output functions. A pin with TRIS clear and ANSEL set will still operate as a digital output, but the Input mode will be analog. This can cause unexpected behavior when executing read-modify-write instructions on the affected port.

REGISTER 3-3: ANSEL: ANALOG SELECT REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
ANS7 ⁽²⁾	ANS6 ⁽²⁾	ANS5 ⁽²⁾	ANS4	ANS3	ANS2	ANS1	ANS0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0

ANS<7:0>: Analog Select bits

Analog select between analog or digital function on pins AN<7:0>, respectively.

1 = Analog input. Pin is assigned as analog input⁽¹⁾.

0 = Digital I/O. Pin is assigned to port or special function.

Note 1: Setting a pin to an analog input automatically disables the digital input circuitry, weak pull-ups, and interrupt-on-change if available. The corresponding TRIS bit must be set to Input mode in order to allow external control of the voltage on the pin.

2: Not implemented on PIC16F883/886.

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3.2.3.8 RA7/OSC1/CLKIN

Figure 3-8 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- a crystal/resonator connection
- a clock input

FIGURE 3-8: BLOCK DIAGRAM OF RA7

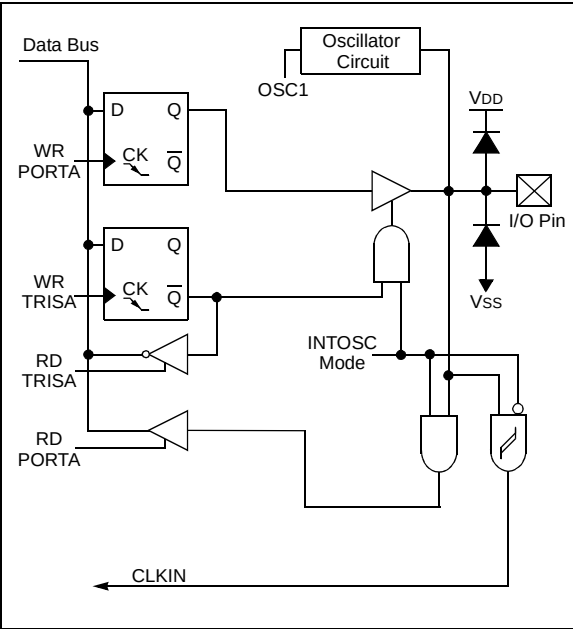


TABLE 3-1: SUMMARY OF REGISTERS ASSOCIATED WITH PORTA

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ADCON0	ADCS1	ADCS0	CHS3	CHS2	CHS1	CHS0	GO/DONE	ADON	0000 0000	0000 0000
ANSEL	ANS7	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0	1111 1111	1111 1111
CM1CON0	C1ON	C1OUT	C1OE	C1POL	—	C1R	C1CH1	C1CH0	0000 -000	0000 -000
CM2CON0	C2ON	C2OUT	C2OE	C2POL	—	C2R	C2CH1	C2CH0	0000 -000	0000 -000
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	0000 --10
PCON	—	—	ULPWUE	SBOREN	—	—	POR	BOR	--01 --qq	--0u --uu
OPTION_REG	RBPu	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111
PORTA	RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0	xxxx xxxx	uuuu uuuu
SSPCON	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	0000 0000	0000 0000
TRISA	TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, — = unimplemented locations read as '0'. Shaded cells are not used by PORTA.

3.3 PORTB and TRISB Registers

PORTB is an 8-bit wide, bidirectional port. The corresponding data direction register is TRISB (Register 3-6). Setting a TRISB bit (= 1) will make the corresponding PORTB pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISB bit (= 0) will make the corresponding PORTB pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-3 shows how to initialize PORTB.

Reading the PORTB register (Register 3-5) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISB register (Register 3-6) controls the PORTB pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISB register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'. Example 3-3 shows how to initialize PORTB.

EXAMPLE 3-3: INITIALIZING PORTB

```
BANKSEL PORTB      ;
CLRF   PORTB        ;Init PORTB
BANKSEL TRISB       ;
MOVLW  B'11110000' ;Set RB<7:4> as inputs
                        ;and RB<3:0> as outputs
MOVWF  TRISB        ;
```

Note: The ANSELH register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

3.4 Additional PORTB Pin Functions

PORTB pins RB<7:0> on the device family device have an interrupt-on-change option and a weak pull-up option. The following three sections describe these PORTB pin functions.

Every PORTB pin on this device family has an interrupt-on-change option and a weak pull-up option.

3.4.1 ANSELH REGISTER

The ANSELH register (Register 3-4) is used to configure the Input mode of an I/O pin to analog. Setting the appropriate ANSELH bit high will cause all digital reads on the pin to be read as '0' and allow analog functions on the pin to operate correctly.

The state of the ANSELH bits has no affect on digital output functions. A pin with TRIS clear and ANSELH set will still operate as a digital output, but the Input mode will be analog. This can cause unexpected behavior when executing read-modify-write instructions on the affected port.

3.4.2 WEAK PULL-UPS

Each of the PORTB pins has an individually configurable internal weak pull-up. Control bits WPUB<7:0> enable or disable each pull-up (see Register 3-7). Each weak pull-up is automatically turned off when the port pin is configured as an output. All pull-ups are disabled on a Power-on Reset by the RBPU bit of the OPTION register.

3.4.3 INTERRUPT-ON-CHANGE

All of the PORTB pins are individually configurable as an interrupt-on-change pin. Control bits IOCB<7:0> enable or disable the interrupt function for each pin. Refer to Register 3-8. The interrupt-on-change feature is disabled on a Power-on Reset.

For enabled interrupt-on-change pins, the present value is compared with the old value latched on the last read of PORTB to determine which bits have changed or mismatched the old value. The 'mismatch' outputs of the last read are OR'd together to set the PORTB Change Interrupt flag bit (RBIF) in the INTCON register.

This interrupt can wake the device from Sleep. The user, in the Interrupt Service Routine, clears the interrupt by:

- Any read or write of PORTB. This will end the mismatch condition.
- Clear the flag bit RBIF.

A mismatch condition will continue to set flag bit RBIF. Reading or writing PORTB will end the mismatch condition and allow flag bit RBIF to be cleared. The latch holding the last read value is not affected by a MCLR nor Brown-out Reset. After these Resets, the RBIF flag will continue to be set if a mismatch is present.

Note: If a change on the I/O pin should occur when the read operation is being executed (start of the Q2 cycle), then the RBIF interrupt flag may not get set. Furthermore, since a read or write on a port affects all bits of that port, care must be taken when using multiple pins in Interrupt-on-Change mode. Changes on one pin may not be seen while servicing changes on another pin.

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REGISTER 3-4: ANSELH: ANALOG SELECT HIGH REGISTER

U-0	U-0	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
—	—	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-6 **Unimplemented:** Read as '0'

bit 5-0 **ANS<13:8>:** Analog Select bits

Analog select between analog or digital function on pins AN<13:8>, respectively.

1 = Analog input. Pin is assigned as analog input⁽¹⁾.

0 = Digital I/O. Pin is assigned to port or special function.

Note 1: Setting a pin to an analog input automatically disables the digital input circuitry, weak pull-ups, and interrupt-on-change if available. The corresponding TRIS bit must be set to Input mode in order to allow external control of the voltage on the pin.

REGISTER 3-5: PORTB: PORTB REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **RB<7:0>:** PORTB I/O Pin bit

1 = Port pin is > V_{IH}

0 = Port pin is < V_{IL}

REGISTER 3-6: TRISB: PORTB TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **TRISB<7:0>:** PORTB Tri-State Control bit

1 = PORTB pin configured as an input (tri-stated)

0 = PORTB pin configured as an output

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REGISTER 3-7: WPUB: WEAK PULL-UP PORTB REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
WPUB7	WPUB6	WPUB5	WPUB4	WPUB3	WPUB2	WPUB1	WPUB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **WPUB<7:0>**: Weak Pull-up Register bit

1 = Pull-up enabled

0 = Pull-up disabled

Note 1: Global $\overline{\text{RBPU}}$ bit of the OPTION register must be cleared for individual pull-ups to be enabled.

2: The weak pull-up device is automatically disabled if the pin is in configured as an output.

REGISTER 3-8: IOCB: INTERRUPT-ON-CHANGE PORTB REGISTER

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
IOCB7	IOCB6	IOCB5	IOCB4	IOCB3	IOCB2	IOCB1	IOCB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **IOCB<7:0>**: Interrupt-on-Change PORTB Control bit

1 = Interrupt-on-change enabled

0 = Interrupt-on-change disabled

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TABLE 3-2: SUMMARY OF REGISTERS ASSOCIATED WITH PORTB

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ANSELH	—	—	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8	--11 1111	--11 1111
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	0000 0000	0000 0000
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	0000 --10
IOCB	IOCB7	IOCB6	IOCB5	IOCB4	IOCB3	IOCB2	IOCB1	IOCB0	0000 0000	0000 0000
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000x
OPTION_REG	RBP $\overline{\text{U}}$	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111
PORTB	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	xxxx xxxx	uuuu uuuu
TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	1111 1111	1111 1111
WPUB	WPUB7	WPUB6	WPUB5	WPUB4	WPUB3	WPUB2	WPUB1	WPUB0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, — = unimplemented read as '0'. Shaded cells are not used by PORTB.

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3.5 PORTC and TRISC Registers

PORTC is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISC (Register 3-10). Setting a TRISC bit (= 1) will make the corresponding PORTC pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISC bit (= 0) will make the corresponding PORTC pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-4 shows how to initialize PORTC.

Reading the PORTC register (Register 3-9) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISC register (Register 3-10) controls the PORTC pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISC register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

EXAMPLE 3-4: INITIALIZING PORTC

```
BANKSEL PORTC      ;
CLRWF  PORTC        ;Init PORTC
BANKSEL TRISC       ;
MOVLW  B'00001100' ;Set RC<3:2> as inputs
MOVWF  TRISC         ;and set RC<7:4,1:0>
                        ;as outputs
```

REGISTER 3-9: PORTC: PORTC REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **RC<7:0>**: PORTC General Purpose I/O Pin bit

1 = Port pin is > V_{IH}

0 = Port pin is < V_{IL}

REGISTER 3-10: TRISC: PORTC TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1 ⁽¹⁾	R/W-1 ⁽¹⁾
TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **TRISC<7:0>**: PORTC Tri-State Control bit

1 = PORTC pin configured as an input (tri-stated)

0 = PORTC pin configured as an output

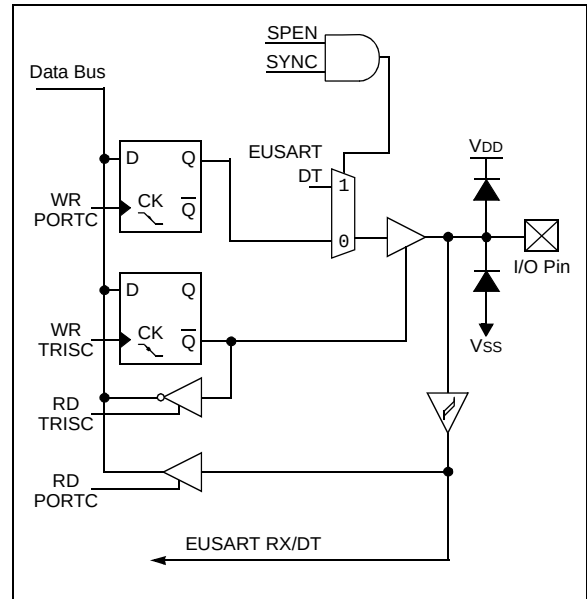
Note 1: TRISC<1:0> always reads '1' in LP Oscillator mode.

3.5.8 RC7/RX/DT

Figure 3-18 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an asynchronous serial input
- a synchronous serial data I/O

FIGURE 3-18: BLOCK DIAGRAM OF RC7



Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	0000 0000	0000 0000
CCP2CON	—	—	DC2B1	DC2B0	CCP2M3	CCP2M2	CCP2M1	CCP2M0	--00 0000	--00 0000
PORTC	RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0	xxxx xxxx	uuuu uuuu
PSTRCON	—	—	—	STRSYNC	STRD	STRC	STRB	STRA	---0 0001	---0 0001
RCSTA	SPEN	RX9	SREN	CREN	ADDEN	FERR	OERR	RX9D	0000 000x	0000 000x
SSPCON	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	0000 0000	0000 0000
T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNCR	TMR1CS	TMR1ON	0000 0000	0000 0000
TRISC	TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, – = unimplemented locations read as '0'. Shaded cells are not used by PORTC.

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3.6 PORTD and TRISD Registers

PORTD⁽¹⁾ is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISD (Register 3-12). Setting a TRISD bit (= 1) will make the corresponding PORTD pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISD bit (= 0) will make the corresponding PORTD pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-5 shows how to initialize PORTD.

Reading the PORTD register (Register 3-11) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

Note 1: PORTD is available on PIC16F884/887 only.

The TRISD register (Register 3-12) controls the PORTD pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISD register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

EXAMPLE 3-5: INITIALIZING PORTD

```
BANKSEL PORTD      ;
CLRF   PORTD        ;Init PORTD
BANKSEL TRISD       ;
MOVLW  B'00001100' ;Set RD<3:2> as inputs
MOVWF  TRISD        ;and set RD<7:4,1:0>
                        ;as outputs
```

REGISTER 3-11: PORTD: PORTD REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RD7	RD6	RD5	RD4	RD3	RD2	RD1	RD0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
-n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **RD<7:0>**: PORTD General Purpose I/O Pin bit
1 = Port pin is > V_{IH}
0 = Port pin is < V_{IL}

REGISTER 3-12: TRISD: PORTD TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISD7	TRISD6	TRISD5	TRISD4	TRISD3	TRISD2	TRISD1	TRISD0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
-n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **TRISD<7:0>**: PORTD Tri-State Control bit
1 = PORTD pin configured as an input (tri-stated)
0 = PORTD pin configured as an output

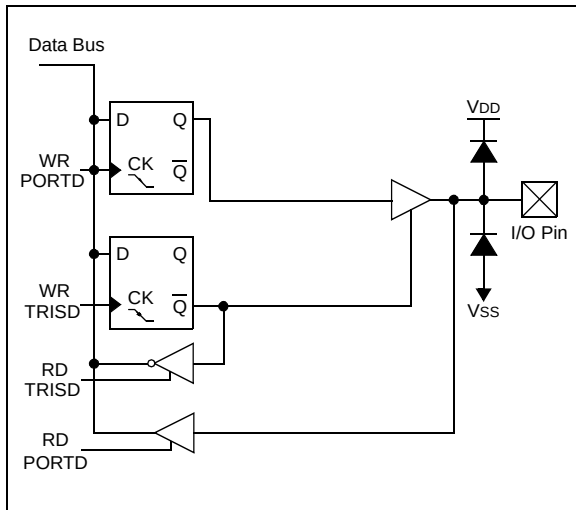
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3.6.1 RD<4:0>

Figure 3-19 shows the diagram for these pins. These pins are configured to function as general purpose I/O's.

Note: RD<4:0> is available on PIC16F884/887 only.

FIGURE 3-19: BLOCK DIAGRAM OF RD<4:0>



3.6.2 RD5/P1B⁽¹⁾

Figure 3-20 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- a PWM output

Note 1: RD5/P1B is available on PIC16F884/887 only. See RB2/AN8/P1B for this function on PIC16F882/883/886.

3.6.3 RD6/P1C⁽¹⁾

Figure 3-20 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- a PWM output

Note 1: RD6/P1C is available on PIC16F884/887 only. See RB1/AN10/P1C/C12IN3- for this function on PIC16F882/883/886.

3.6.4 RD7/P1D⁽¹⁾

Figure 3-20 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- a PWM output

Note 1: RD7/P1D is available on PIC16F884/887 only. See RB4/AN11/P1D for this function on PIC16F882/883/886.

FIGURE 3-20: BLOCK DIAGRAM OF RD<7:5>

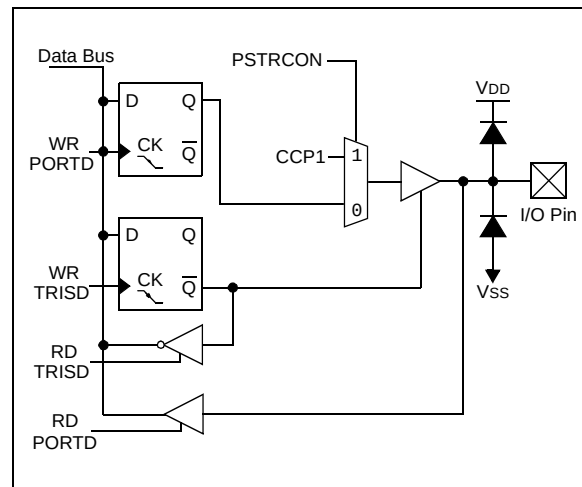


TABLE 3-4: SUMMARY OF REGISTERS ASSOCIATED WITH PORTD

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
PORTD	RD7	RD6	RD5	RD4	RD3	RD2	RD1	RD0	xxxx xxxx	uuuu uuuu
PSTRCON	—	—	—	STRSYNC	STRD	STRC	STRB	STRA	---0 0001	---0 0001
TRISD	TRISD7	TRISD6	TRISD5	TRISD4	TRISD3	TRISD2	TRISD1	TRISD0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, — = unimplemented locations read as '0'. Shaded cells are not used by PORTD.

3.7 PORTE and TRISE Registers

PORTE⁽¹⁾ is a 4-bit wide, bidirectional port. The corresponding data direction register is TRISE. Setting a TRISE bit (= 1) will make the corresponding PORTE pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISE bit (= 0) will make the corresponding PORTE pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). The exception is RE3, which is input only and its TRIS bit will always read as '1'. Example 3-6 shows how to initialize PORTE.

Reading the PORTE register (Register 3-13) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch. RE3 reads '0' when MCLR = 1.

Note 1: RE<2:0> pins are available on PIC16F884/887 only.

The TRISE register (Register 3-14) controls the PORTE pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISE register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

Note: The ANSEL register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

EXAMPLE 3-6: INITIALIZING PORTE

```
BANKSEL PORTE      ;
CLRF   PORTE       ;Init PORTE
BANKSEL ANSEL      ;
CLRF   ANSEL       ;digital I/O
BCF    STATUS,RP1  ;Bank 1
BANKSEL TRISE      ;
MOVLW  B'00001100' ;Set RE<3:2> as inputs
MOVWF  TRISE       ;and set RE<1:0>
                        ;as outputs
```

REGISTER 3-13: PORTE: PORTE REGISTER

U-0	U-0	U-0	U-0	R-x	R/W-x	R/W-x	R/W-x
—	—	—	—	RE3	RE2	RE1	RE0
bit 7				bit 0			

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-4 **Unimplemented:** Read as '0'
 bit 3-0 **RD<3:0>:** PORTE General Purpose I/O Pin bit
 1 = Port pin is > V_{IH}
 0 = Port pin is < V_{IL}

REGISTER 3-14: TRISE: PORTE TRI-STATE REGISTER

U-0	U-0	U-0	U-0	R-1 ⁽¹⁾	R/W-1	R/W-1	R/W-1
—	—	—	—	TRISE3	TRISE2	TRISE1	TRISE0
bit 7				bit 0			

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-4 **Unimplemented:** Read as '0'
 bit 3-0 **TRISE<3:0>:** PORTE Tri-State Control bit
 1 = PORTE pin configured as an input (tri-stated)
 0 = PORTE pin configured as an output

Note 1: TRISE<3> always reads '1'.

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3.7.1 RE0/AN5⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE0/AN5 is available on PIC16F884/887 only.

3.7.2 RE1/AN6⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE1/AN6 is available on PIC16F884/887 only.

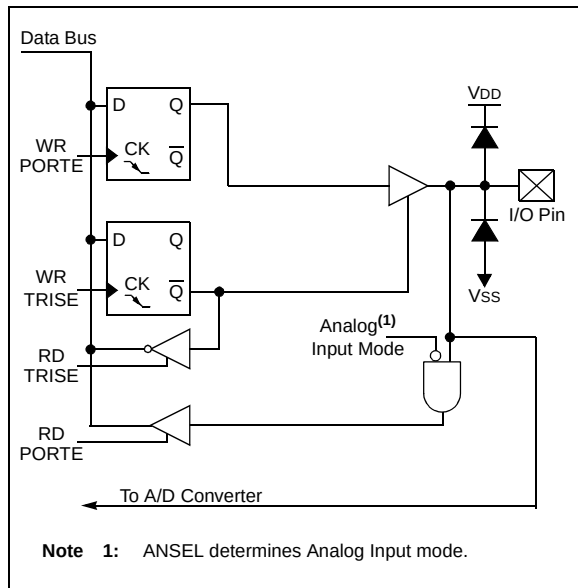
3.7.3 RE2/AN7⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE2/AN7 is available on PIC16F884/887 only.

FIGURE 3-21: BLOCK DIAGRAM OF RE<2:0>



3.7.4 RE3/MCLR/VPP

Figure 3-22 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose input
- as Master Clear Reset with weak pull-up

FIGURE 3-22: BLOCK DIAGRAM OF RE3

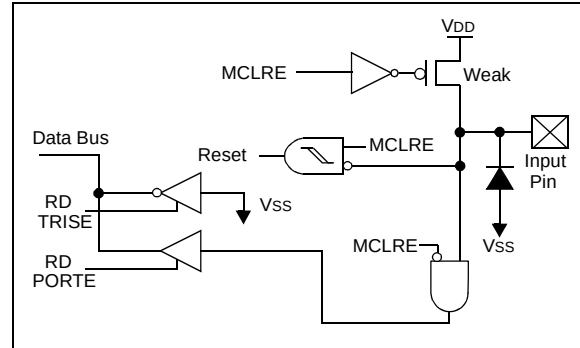


TABLE 3-5: SUMMARY OF REGISTERS ASSOCIATED WITH PORTE

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ANSEL	ANS7	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0	1111 1111	1111 1111
PORTE	—	—	—	—	RE3	RE2	RE1	RE0	---- xxxx	---- uuuu
TRISE	—	—	—	—	TRISE3	TRISE2	TRISE1	TRISE0	---- 1111	---- 1111

Legend: x = unknown, u = unchanged, – = unimplemented locations read as '0'. Shaded cells are not used by PORTE